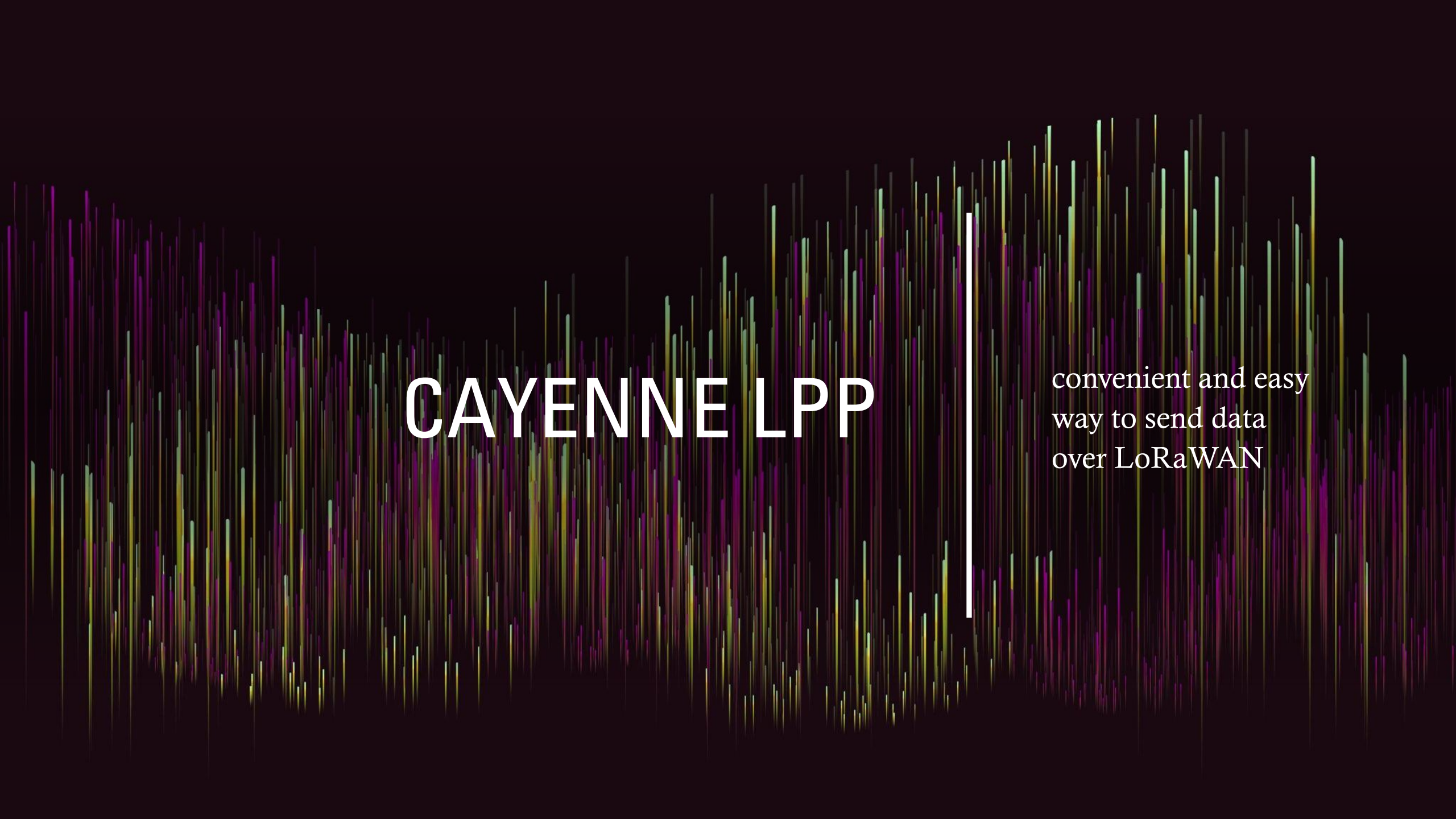
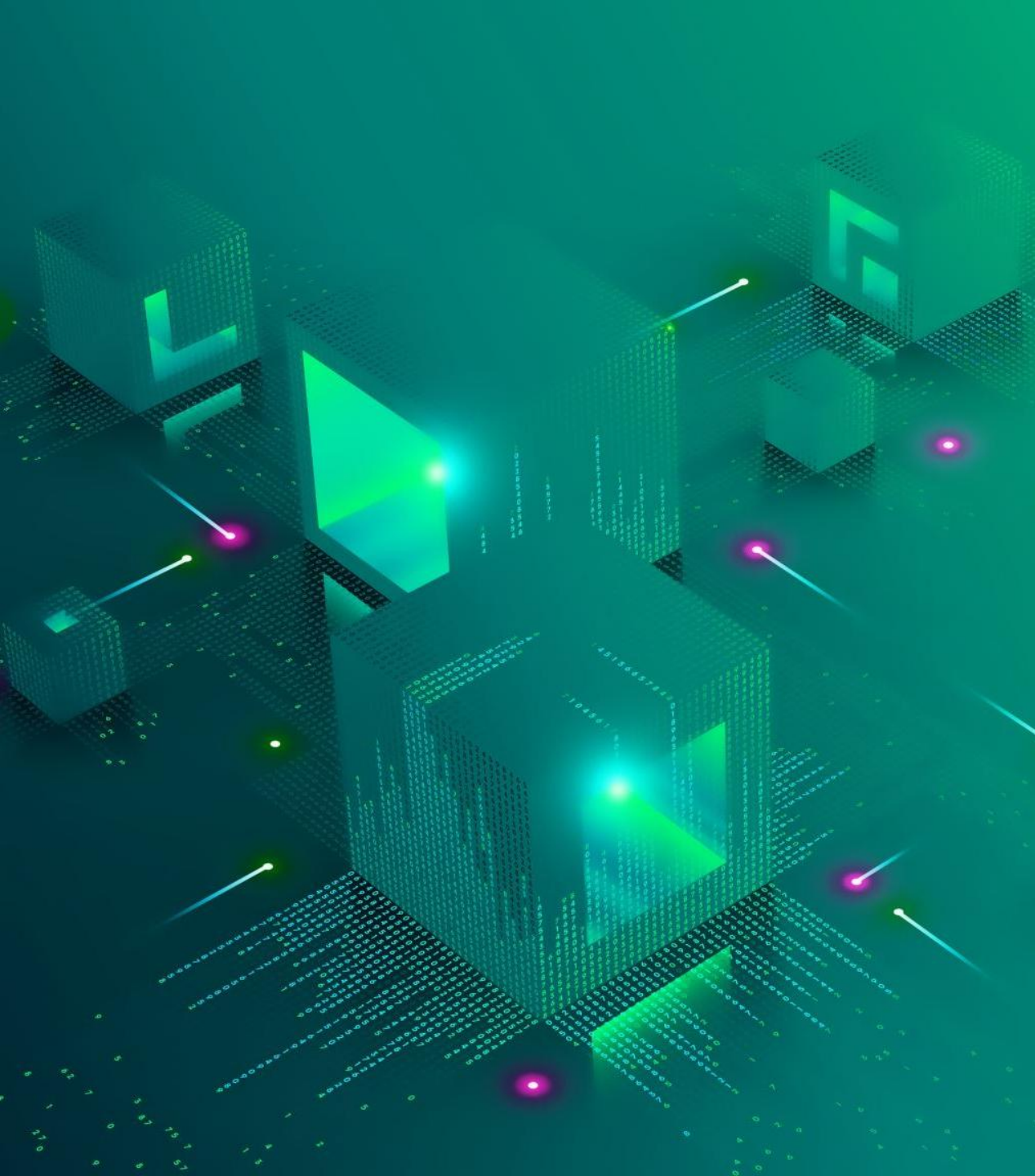


The background of the slide is a dark purple color. It is covered with a dense field of vertical lines of varying heights and colors, including shades of purple, green, and yellow. These lines create a textured, digital-like effect. In the center of the slide, the text 'CAYENNE LPP' is displayed in a large, white, sans-serif font. To the right of this text, there is a thin, vertical white line that extends from the top of the text area down towards the bottom of the slide. Further to the right, there is a block of text in a smaller, white, sans-serif font.

CAYENNE LPP

convenient and easy
way to send data
over LoRaWAN



CAYENNE LPP

- Cayenne Low Power Payload (LPP) is a data format designed by myDevices for sending sensor data efficiently over low-power IoT networks like LoRaWAN.
 - It defines a compact binary encoding for sensor readings (temperature, humidity, GPS, etc.), so that microcontrollers with limited bandwidth and power can send data easily.
 - You can find the official documentation her:
<https://docs.mydevices.com/docs/lorawan/cayenne-lpp>
-

PAYLOAD STRUCTURE

1 Byte	1 Byte	N Bytes	1 Byte	1 Byte	M Bytes	...
Data1 Channel	Data1 Type	Data1	Data2 Channel	Data2 Type	Data 2	...



Channel: identifies which sensor
(e.g. 1 = sensor on pin 8)



Data Type: tells what kind of data follows
(e.g. 0x67 = temperature)



Data: the actual measurement in a standardized, fixed-length format

PREDEFINED DATA TYPES

Type	LPP	Hex	Data Size	Data Resolution per bit
Digital Input	0	0	1	1
Digital Output	1	1	1	1
Analog Input	2	2	2	0.01
Analog Output	3	3	2	0.01
Illuminance Sensor	101	65	2	1
Presence Sensor	102	66	1	1
Temperature Sensor	103	67	2	0.1°C Signed MSB

Type	LPP	Hex	Data Size	Data Resolution per bit
Humidity Sensor	104	68	1	0.5 % Unsigned
Accelerometer	113	71	6	0.001 G Signed MSB per axis
Barometer	115	73	2	0.1 hPa Unsigned MSB
Gyrometer	134	86	6	0.01 °/s Signed MSB per axis
GPS Location	136	88	9	Latitude / Longitude : 0.0001 ° Signed MSB
GPS Location	136	88	9	Altitude : 0.01 meter Signed MSB

ARDUINO IMPLEMENTATION

```
class CayenneLPP {
public:
    CayenneLPP(uint8_t size);
    ~CayenneLPP();

    void reset(void);
    uint8_t getSize(void);
    uint8_t* getBuffer(void);
    uint8_t copy(uint8_t* buffer);

    uint8_t addDigitalInput(uint8_t channel, uint8_t
        value);
    uint8_t addDigitalOutput(uint8_t channel, uint8_t
        value);

    uint8_t addAnalogInput(uint8_t channel, float value);
    uint8_t addAnalogOutput(uint8_t channel, float value);
```

```
    uint8_t addLuminosity(uint8_t channel, uint16_t lux);
    uint8_t addPresence(uint8_t channel, uint8_t value);
    uint8_t addTemperature(uint8_t channel, float celsius);
    uint8_t addRelativeHumidity(uint8_t channel, float rh);
    uint8_t addAccelerometer(uint8_t channel, float x,
        float y, float z);
    uint8_t addBarometricPressure(uint8_t channel, float
        hpa);
    uint8_t addGyrometer(uint8_t channel, float x, float y,
        float z);
    uint8_t addGPS(uint8_t channel, float latitude, float
        longitude, float meters);

private:
    uint8_t *buffer;
    uint8_t maxsize;
    uint8_t cursor;

};
```

WHY USING CAYENNE LPP



Easy integration with IoT platforms (like The Things Network)



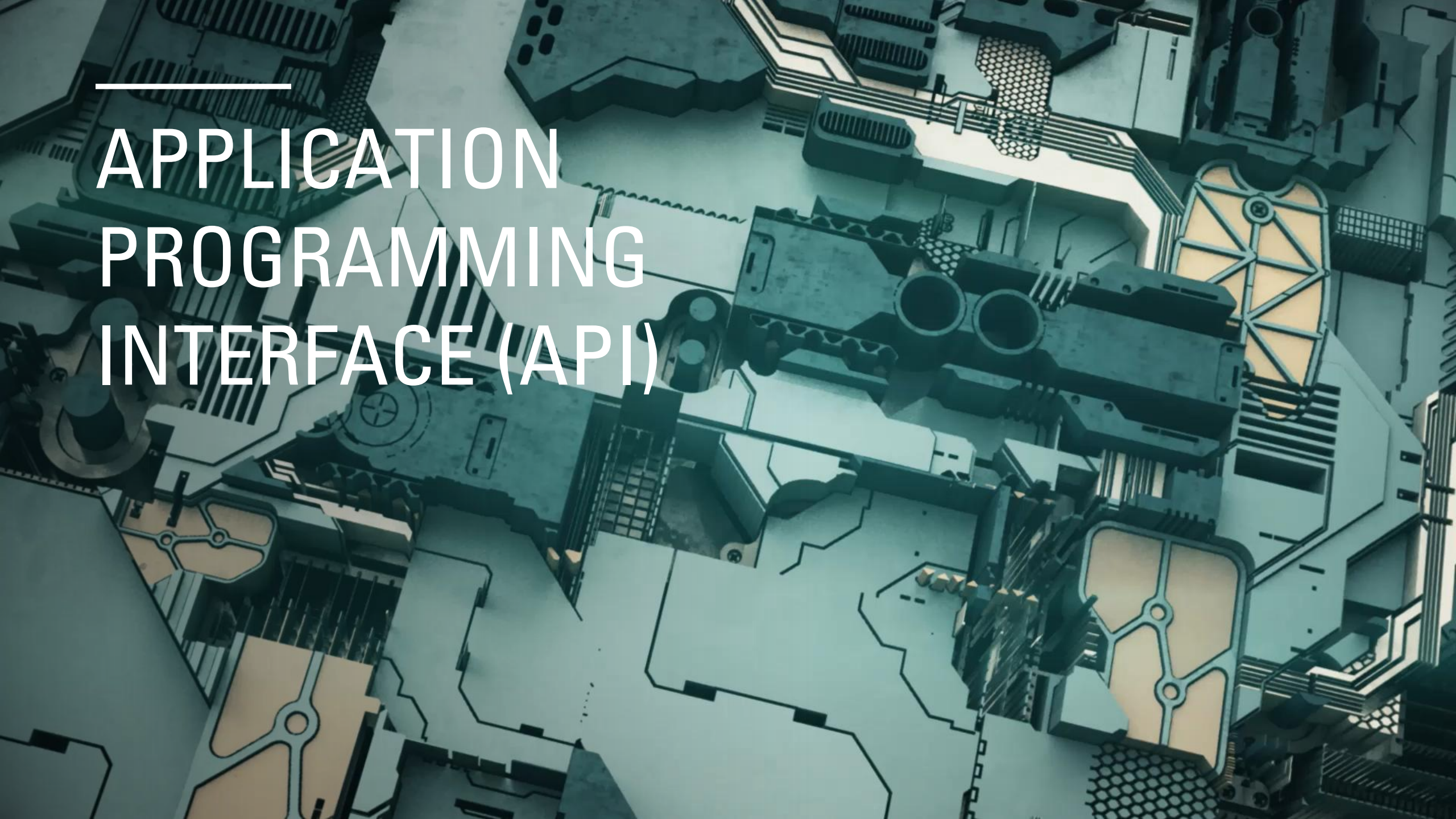
Saves bandwidth and power



Standardized data types and scaling factors



Readable across many devices and languages



APPLICATION PROGRAMMING INTERFACE (API)

WHAT IS AN API?

It's a set of rules that lets two applications talk to each other.

For example:

- Your IoT device sends data to The Things Network (TTN).
- Another app can get that data using TTN's API.





WHAT IS AN API KEY?

An API key is like a password for your application. It identifies who is making the request and what they are allowed to do.

In TTN, each application or user gets a unique key.

The key gives permission to:

- Read uplink data from devices
- Send downlink messages
- Manage devices or applications
- ...

WITHOUT AN API KEY



ANYONE COULD ACCESS
YOUR SENSOR DATA



DEVICES COULD BE
CONTROLLED BY
STRANGERS



NETWORK RESOURCES
COULD BE ABUSED



IT GRANTS



Authentication

proves you are allowed to access data



Authorization

limits *what* you can do (e.g., only read data)



Security

keeps your devices and network safe
